Representing Data

The datasets we used in all three charts track public funding for higher education. In particular, it measures funding in dollars per Full-Time Equivalent (FTE) student over 30 years, from 1990-91 to 2020-21. Showing data in a chart helps users understand the overall patterns much faster than reading a text list of numbers. Each chart changes how this data is shown to highlight a different aspect of the information. The chart titled "*30-Year Trend of State Appropriations for Higher Education By Selected State*" is a line graph that shows how funding changes over the full 30 years for five specific states: California, New York, Texas, Florida, and Illinois. The second chart, titled "*State Higher Education Funding 2020-21*," focuses on a single point in time by filtering for the most recent academic year from 2020-21. It shows the funding levels of the top ten states side-by-side and includes a national average line. The dynamic dashboard featuring panels titled "*Percent Change in State Appropriations (1990-91 to 2020-21)*" and "*Distribution of State Appropriations Over Time*" shows the total percentage change in funding from the first year to the last year while also showing how the data is spread out across all of the years.

Key Characteristics

Each chart uses different layouts and design choices to display the values. "*30-Year Trend of State Appropriations for Higher Education By Selected State*" is a line graph with clear markers at each data point. A line chart connects data points with straight lines to show a trend, making it perfect for tracking changes over time on a horizontal axis. The layout features five different colored lines on a grid with a light-gray background. The horizontal X-axis shows the academic years turned at a 45-degree angle so they are easy to read. The vertical Y-axis tracks the funding scale, and a legend that matches each state to its colored line.

The chart titled "*State Higher Education Funding 2020-21*" is a simple vertical bar chart with skyblue bars to keep the design clean. A bar chart uses rectangular bars to show data for different categories, where the length of the bar matches the value of that category. It places the bars side-by-side, and a horizontal red dashed line cuts across the chart to mark a baseline. The horizontal X-axis lists the state names turned at 45 degrees, and the vertical Y-axis measures the funding amount. A legend labels the read line as the national average.

The two-panel dashboard uses a composite, stacked layout. The top panel, titled "*Percent Change in State Appropriations (1990-91 to 2020-21)*," is a horizontal bar chart where the length of each bar shows the value, splitting left and right from a vertical line at zero percent. The bottom panel, titled "*Distribution of State Appropriations Over Time,*" shows a row of box-and-whisker plots. The top panel puts states on the vertical axis and percentages on the horizontal axis, using red for a drop in funding and green for funding growth. The bottom panel puts time intervals on the horizontal axis and the funding spread on the vertical axis, showing the middle values, quartiles, and outliers.

Comparative Evaluation

Choosing the right graph depends on matching the chart type to what you want to say about the data. "*30-Year Trend of State Appropriations for Higher Education By Selected States*" is excellent for showing long-term trends and how fast values change over time. Limiting the chart to five states keeps it clean, making it easy to follow each line and see major drops, like the funding dip around 2010-11. Line charts fit continuous time data perfectly because the connecting lines show changes from one year to the next.

The chart titled "*State Higher Education Funding 2020-21*" gives great clarity for ranking different states. The vertical bars make it easy to compare heights, and the average line tells you immediately which states are above the national average and which are below it. It aligns well with data rules by using a continuous vertical scale for the money values and separate bars for independent categories like states.

The dashboard containing "*Percent Change in State Appropriations (1990-91 to 2020-21)*" and "*Distribution of State Appropriations Over Time*" provides deep statistical details but takes more effort to read. The top panel uses red and green bars to show funding growth versus decline instantly. The horizontal layout works well here because it gives plenty of room for the long state names on the vertical axis. The box plots in the bottom panel are great for showing how the data spreads out over several decades, but this panel requires more data experience to read than the simple lines and bars of other charts.

Audience Selection and Rationale

When presenting to executive leaders, like policy officials or budget committees, the main goal is to help them make quick decisions. Visualizations guides emphasize that you must match your chart choices to the time constraints of your audience (Atlassian, n.d.). For this busy audience, "*State Higher Education Funding 2020-21*" (the simple vertical bar chart) is the best choice. Executive leaders rarely have the time to study complex data spreads or untangle intersecting lines. They need quick answers to simple questions like: Where do we stand today, who is leading, and are we above or below the national average? This kind of chart gives the answers immediately. It groups current data, ranks performance by bar height, and uses the average line as a clear yardstick. It tells a simple story that requires no advanced data training to understand.

Technical Refinements Recommendations

Shander (2016) noted that a few simple design changes can make these charts much easier for an executive audience to read.

First, for “*30-Year Trend of State Appropriations for Higher Education By Selected States,*” the numbers on the vertical Y-axis should include a dollar sign (“$”), and the scale should start exactly at zero instead of starting from a cutoff. Adding currency symbols removes any confusion about the values. Starting the axis at zero ensures the chart stays accurate, preventing small funding drops from looking like huge collapses.

Second, for “*State Higher Education Funding 2020-21,*” the bars should be sorted in order from the highest funding to the lowest, and the exact dollar values should be printed right on the top of each bar. Sorting by volume lets busy leaders spot top spenders and low outliers instantly. Adding exact numbers saves the reader from guessing values by tracing a line to a distant axis.

Finally, for the dashboard featuring “*Percent Change in State Appropriations (1990-91 to 2020-21),*” the red and green colors should be replaced with a color-blind accessible option, like dark blue for growth and bright orange for a drop. Red-green colorblindness is the most common form of color vision deficiency. Using a high-contrast blue and orange palette ensures that important trends stay readable for everyone in your audience.

References

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